Worksheet 6.1 — Compound Angle Formulas

Grade 12 Advanced Functions (MHF4U) · Unit 6: Trigonometric Identities and Equations

The Big Ideas

$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ and $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$ give exact values for new angles.

- $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$.
- $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$.
- Write $15°, 75°, \ldots$ as sums/differences of special angles.

Core Concepts

Angles added together

The **compound angle** formulas express $\sin(A \pm B)$ and $\cos(A \pm B)$ using the ratios of A and B separately.

The key identities

$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ and $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$.

Exact new angles

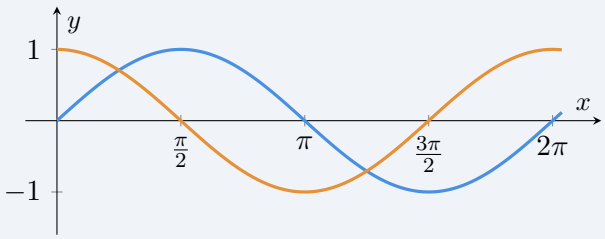
They give exact values for angles like $75° = 45° + 30°$ from the known special angles.

Worked Examples

Example 1 — Cosine of a sum

Find $\cos 345° = \cos(300° + 45°)$.

Solution. $\cos 300 \cos 45 - \sin 300 \sin 45 = \frac{1}{2} \cdot \frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2} + \sqrt{6}}{4}$. Recall cos is sin shifted by $\frac{\pi}{2}$:



Example 2 — Sine of a sum

Find $\sin 285° = \sin(240° + 45°)$.

Solution. $\sin 240 \cos 45 + \cos 240 \sin 45 = -\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} - \frac{1}{2} \cdot \frac{\sqrt{2}}{2} = -\frac{\sqrt{6} + \sqrt{2}}{4}$.

Example 3 — Difference

Find $\sin 15^\circ = \sin(45^\circ - 30^\circ)$.

Solution. $\sin 45 \cos 30 - \cos 45 \sin 30 = \dfrac{\sqrt{6}-\sqrt{2}}{4}$.

Example 4 — Recognize the pattern

Simplify $\cos x \cos y + \sin x \sin y$.

Solution. This is $\cos(x - y)$.

Example 5 — Use given values

$\sin A = \frac{8}{17}, \cos A = \frac{15}{17}, \sin B = \frac{4}{5}, \cos B = \frac{3}{5}$. Find $\sin(A + B)$.

Solution. $\frac{8}{17} \cdot \frac{3}{5} + \frac{15}{17} \cdot \frac{4}{5} = \frac{24+60}{85} = \dfrac{84}{85}$.

Example 6 — Tangent difference

Find $\tan 15^\circ = \tan(45^\circ - 30^\circ)$.

Solution. $\dfrac{\tan 45 - \tan 30}{1 + \tan 45 \tan 30} = \dfrac{1 - 1/\sqrt{3}}{1 + 1/\sqrt{3}} = \dfrac{\sqrt{3}-1}{\sqrt{3}+1} = 2 - \sqrt{3}$.

Example 7 — Recognize

Simplify $\sin x \cos y + \cos x \sin y$.

Solution. $\sin(x + y)$.

Example 8 — Recognize

Simplify $\cos x \cos y - \sin x \sin y$.

Solution. $\cos(x + y)$.

Example 9 — Use given values

With the Example-5 values, find $\cos(A + B)$.

Solution. $\frac{15}{17} \cdot \frac{3}{5} - \frac{8}{17} \cdot \frac{4}{5} = \frac{45-32}{85} = \dfrac{13}{85}$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Find $\sin 345°$.

Question 2

Simplify $\sin 3x \cos x + \cos 3x \sin x$.

Question 3

Simplify $\cos 5x \cos 2x - \sin 5x \sin 2x$.

Question 4

Find $\cos 285° = \cos(240° + 45°)$.

Question 5

With Example-5 values, find $\sin(A - B)$.

Question 6

Find $\cos 105° = \cos(60° + 45°)$.

Question 7

Simplify $\cos 40° \cos 10° + \sin 40° \sin 10°$.

Question 8

State the formula for $\sin(A - B)$.

Question 9

Find $\tan 75° = \tan(45° + 30°)$.

Question 10

State the formula for $\cos(A+B)$.

Question 11

Simplify $\sin 80^\circ \cos 20^\circ - \cos 80^\circ \sin 20^\circ$.

Question 12

Find $\tan(A-B)$ given $\tan A = 3, \tan B = 2$.

Question 13 — Challenge

Use $\sin A = \frac{7}{25}, \cos A = \frac{24}{25}, \sin B = \frac{12}{13}, \cos B = \frac{5}{13}$ to find $\sin(A-B)$.

Question Recap & Answer Key

Question Recap

1. Find $\sin 345°$.
2. Simplify $\sin 3x \cos x + \cos 3x \sin x$.
3. Simplify $\cos 5x \cos 2x - \sin 5x \sin 2x$.
4. Find $\cos 285° = \cos(240° + 45°)$.
5. With Example-5 values, find $\sin(A - B)$.
6. Find $\cos 105° = \cos(60° + 45°)$.
7. Simplify $\cos 40° \cos 10° + \sin 40° \sin 10°$.
8. State the formula for $\sin(A - B)$.
9. Find $\tan 75° = \tan(45° + 30°)$.
10. State the formula for $\cos(A + B)$.
11. Simplify $\sin 80° \cos 20° - \cos 80° \sin 20°$.
12. Find $\tan(A - B)$ given $\tan A = 3, \tan B = 2$.
13. Use $\sin A = \frac{7}{25}, \cos A = \frac{24}{25}, \sin B = \frac{12}{13}, \cos B = \frac{5}{13}$ to find $\sin(A - B)$.

✓ Answer Key

1. $\frac{\sqrt{2}-\sqrt{6}}{4}$
2. $\sin 4x$
3. $\cos 7x$
4. $\frac{\sqrt{6}-\sqrt{2}}{4}$
5. $-\frac{36}{85}$
6. $\frac{\sqrt{2}-\sqrt{6}}{4}$
7. $\cos 30° = \frac{\sqrt{3}}{2}$
8. $\sin A \cos B - \cos A \sin B$
9. $2 + \sqrt{3}$
10. $\cos A \cos B - \sin A \sin B$
11. $\sin 60° = \frac{\sqrt{3}}{2}$
12. $\frac{3-2}{1+6} = \frac{1}{7}$
13. $\frac{35-288}{325} = -\frac{253}{325}$